

GRIDBENCH · REQUIREMENTS

GridBench requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gridbench		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-02	Amish Chadha	R7 note: light-duty rating and marking at the 122 screw-in positions decided by Amish on 2026-10-02; R7 unchanged
0.5	2026-10-01	Amish Chadha	Design made constructable (GBN-DDR-003); R10 reported against the value-engineering target; R12 figures updated
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from GBN-CAL-001; R10 redefined to exclude the user-supplied indicator and R12 assessed with ballast (GBN-DDR-001)
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (12 measurable requirements, status against the concept)
0.1	2026-09-25	Amish Chadha	Initial scaffold

GridBench requirements

Twelve requirements define a garage-built grid bench and fixture set. The TRL 3 calculations (GBN-CAL-001 v0.3) find one not met in part: clamp hold-down (R7) at the 122 insert positions that sit over a frame member, where a screw-in insert may pull out. Two are at risk (R3, R4), one cannot be verified on paper (R5), four are met on paper and three are met by design. R10 is reported against the value-engineering target (Amish, 2026-10-01): value-engineering target USD 250; estimated cost of the constructable design USD 268.40 (USD 18.40 over the target), after the joints, fixings and clamps that make the design buildable (GBN-DDR-003). At v0.4, under GBN-DDR-002, the R10 target became \$250, the `budget_usd` Amish set on 2026-09-25. At v0.3, under GBN-DDR-001, R10 was redefined to exclude the user-supplied dial indicator and R12 is assessed with the shelf ballast in place.

■ Requirements

Table 1. GridBench requirements and TRL 3 status (GBN-CAL-001, Table 2).

ID	Requirement	Target	Verification	TRL 3 status
R1	Grid standard	25.00 mm pitch in X and Y, M6 threads, hole centers 12.5 mm from the worktop edges; fits metric optical breadboard accessories	Inspection of drawing; fit test with a commercial M6 accessory	Met by design: 1,152 positions, tile on the same grid
R2	Worktop size and height	At least 1,200 x 600 mm; top at 900 mm, adjustable ± 15 mm with levelling feet	Measurement	Met by design
R3	Coarse field hole position	± 0.3 mm hole to hole over 1,000 mm at 40 to 60 % RH (template-drilled build: ± 0.5 mm)	Measurement with steel rule and pin gauges	At risk: CNC ± 0.22 mm; template ± 0.34 mm statistical, ± 0.62 mm worst case; moisture coefficient unsourced
R4	Precision tile location	Hole position ± 0.05 mm; a part relocated on two 8 mm dowels (one round, one diamond) returns within 0.03 mm	Dial indicator test, 10 relocations	At risk: 0.036 mm worst case, 0.015 mm statistical; hole position not verifiable at TRL 3
R5	Flatness	Tile 0.05 mm over 300 mm; worktop 0.5 mm over 1,000 mm	Straightedge and feeler gauges	Not verifiable at TRL 3 (depends on the plate and sheet as supplied)
R6	Stiffness and load	150 kg distributed load; deflection 0.5 mm or less under 500 N at the middle of a bay	Load test	Met on paper: 0.37 mm with 45 x 120 mm aprons; 150 kg at a sixteenth of the timber strength
R7	Clamp hold-down	Each printed toe clamp holds 500 N at the part with no visible creep after 8 h at 25 °C	Spring scale pull test	Not met at the 122 screw-in positions over the frame (pull-out factor 0.57 to 0.92); met on paper at the 130 tee-nut positions (factor 1.51 to 2.42) and on the tile (515 to 858 N at 1.6 N·m, 9.9 MPa)
R8	Fixture change	Swap one fixture for another in 60 s or less with one 5 mm hex key	Timed trial	Met on paper: 56 s (thin margin)
R9	Buildability	Built with hand and garage tools plus an FDM printer (200 x 200 mm bed); CNC optional	Build review	Met on paper: largest print 190 mm; tile about 4.2 h by drill press
R10	Parts cost	Value-engineering target USD 250 (<code>budget_usd</code> , a hypothetical control target) for bench, grid, tile and fixture set; the dial indicator is user-supplied and excluded (GBN-DDR-001 A1)	Priced BOM	Over the value-engineering target by \$18.40: \$268.40 for the constructable design (GBN-DDR-003)
R11	Openness	All geometry as build123d source; fixture library parametric; CERN-OHL-S-2.0	Repository review	Met by design
R12	Stability and edges	No tipping under 150 N horizontal at the top edge, with the shelf ballast in place; all exposed edges chamfered or rounded 0.5 mm or more	Push test; inspection	Met on paper with 25.8 kg of ballast (186 N); the empty bench tips at 117 N

■ Design load case

- Distributed load: 150 kg over the worktop (a fixture set, tools and parts).
- Point load: 500 N from a person leaning on a clamped part, at the middle of a bay between cross rails.
- Horizontal push: 150 N at the front edge of the worktop, as when pushing a tight part onto dowels or filing.

■ Assumptions

- Birch plywood with a modulus of about 7 GPa, mean of the two in-plane directions (still an assumption; GBN-CAL-001, Table 1).
- C16 softwood frame at about 500 kg/m³, mean modulus 8 GPa.
- Workshop humidity between 40 and 60 % RH for R3, a moisture content swing of ± 1.65 % (USDA *Wood Handbook*, Table 4-2).

■ Requirements not met

- **R7 clamp hold-down at the 122 screw-in positions over the frame:** the screw tension that guarantees 500 N at the part can exceed the estimated pull-out strength of a screw-in insert. Tee nuts, decided under GBN-DDR-002, cannot be fitted there because a frame member blocks the underside. Decided by Amish on 2026-10-02 (GBN-DDR-002, 04): these positions are rated light duty at 0.92 N·m (296 to 493 N at the part), marked on the worktop in a contrasting colour and stated in the fixture library notes. R7 itself is unchanged and stays not met at these positions.
- **R3 and R4 at risk:** see GBN-CAL-001, sections E and F. The precision tile carries all tight-tolerance work.